Hitotsubashi University, Graduate School of Economics

International Development Economics II

Instructor: Kazuki Motohashi

Fall-Winter 2026
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Course Information

- **Classes:** Tuesday 13:15-15:00
- **Office hours:** Tuesday 12:00-13:00 (By appointment. If you would like to attend office hours, please contact the below email address by the day before.)
- **Contact:** kazuki.motohashi@r.hit-u.ac.jp

Course Description

This course explores advanced topics at the intersection of development and environmental economics. In the first part of the course, we will review fundamental theories of environmental economics, causal inference methods, and geographic information system (GIS) data and analysis. We will then examine recent economics papers analyzing environmental policy issues in developing countries. By the end of the course, students will develop research proposals on relevant topics.

Learning Objectives

In this course, students will:

1. Learn microeconomic theories and empirical methodologies relevant to analyzing environmental policy issues in developing countries.
2. Explore recent applied literature at the intersection of development and environmental economics.
3. Develop their own research proposals related to development or environmental economics.

Prerequisites

This course is designed for graduate students interested in development economics, environmental economics, or related fields. Students are expected to have completed upper-level undergraduate or graduate courses in microeconomics and econometrics as prerequisites.

Course Materials

Reference (Optional) Textbook for Class 1
Daniel Phaneuf and Till Requate. *A Course in Environmental Economics: Theory, Policy, and Practice*. Cambridge University Press, 2016.

Reference (Optional) Textbook for Class 2
Scott Cunningham. *Causal Inference: The Remix*. Yale University Press, 2026.
An online version is also available [link]

Reference Website for Class 3
R as GIS for Economists [link]

Lecture Slides for Classes 1–6 & 8–11

Course Requirements

Referee Reports

Starting from Class 4, students are required to read the assigned papers (two per class) and prepare brief referee reports (about 400–500 words) for at least one paper in each class. The referee reports must address the following points:

- Brief summary of the paper (e.g., research question, methodology, results)
- Why is the paper important (or not)?
- What did you like or dislike about the paper? What could the authors have done differently?
- Which parts of the paper were unclear or needed further explanation?
- What potential research ideas could you develop based on this paper?

Paper Presentations

Students assigned to present the papers (one student per paper, 20 minutes per paper) are required to prepare slides in PowerPoint or Beamer. The presentation must address the following points:

- Motivation (problem and contribution)
- Research question
- Conceptual framework or theoretical model (if any)
- Study context
- Data
- Empirical strategy (including tests of identifying assumptions)
- Main results (citing main tables and figures)

Both referee reports and presentation slides are due the day before class and should be posted on the Manaba Forum. The instructor will create a discussion thread for each class, so please post your materials in the designated thread.

Research Proposal

Students are also required to develop research proposals related to development or environmental economics, preferably at their intersection. They must prepare a midterm presentation, a final presentation, and a final submission of their research proposals. For second-year master's students, a draft of their master's thesis may be used to fulfill this requirement. The presentation duration and the expected length of the final paper will be determined based on the number of enrolled students.

The presentations and final paper must include the following parts:

- Motivation (problem, key related literature, and contribution)
- Research question
- Conceptual framework or theoretical model (if any)
- Study context
- Data
- Empirical strategy (including tests of identifying assumptions)
- Preliminary results (optional)

Please feel free to attend office hours to discuss your ideas or the paper with me at any stage of your research. If needed, students may also schedule a meeting with me outside of office hours.

Grading

Evaluation will be based on the following three components:

1. Presentation and referee reports on assigned papers (40%)
2. Contribution to in-class discussions (30%)
3. Research proposal presentation and final paper (30%)

Course Schedule and Reading List

This course will be conducted in a seminar format. Students are required to read assigned papers, give presentations, and actively participate in discussions. Presentation assignments will be determined in Classes 1 and 2, so please make sure to attend. Moreover, starting in Class 4, the instructor will provide a brief overview of each class topic and cover research skills modules designed to support students' research more broadly.

Class 1 (Sep 15): Course Introduction and Theories of Externalities and Pollution Control Instruments

Required readings: Greenstone and Jack (2015)
Optional readings: Phaneuf and Requate, Chapters 1–3; Jayachandran (2022); Jack and Ryan (2026)

Class 2 (Sep 29): Causal Inference Methods (RCT, IV, RDD, DiD)

Optional readings: Cunningham, Chapters 4, 6, 7, 9

Class 3 (Oct 6): GIS Data and Analysis

Required readings: R as GIS for Economists, Chapter 1

Class 4 (Oct 13): Climate Change and Adaptation

Required readings: Dell et al. (2012); Colmer (2021)
Optional readings: Liu et al. (2023); Somanathan et al. (2021); Motohashi (2026a)
Research skill module: Environmental Valuation (Discrete Choice Models and Hedonic Methods)

Class 5 (Oct 20): Environmental Pollution and Health Impacts

Required readings: Garg et al. (2024); Do et al. (2018)
Optional readings: Dias et al. (2023); Rangel and Vogl (2019); Tanaka et al. (2022)
Research skill module: Tips for Generating Research Ideas

Class 6 (Oct 27): Regulation-Based Approaches to Pollution Control

Required readings: Duflo et al. (2013); He et al. (2020)
Optional readings: Greenstone and Hanna (2014); Buntaine et al. (2024); Assunção et al. (2023); Axbard and Deng (2024); Iqbal et al. (2024)
Research skill module: Presentation Tips

Class 7 (Nov 3): Midterm Presentation of Research Proposals

Class 8 (Nov 10): Market-Based Approaches to Pollution Control

Required readings: Greenstone et al. (2025); Davis et al. (2014)
Optional readings: Jack et al. (2025); Calel et al. (2025); Li and Zhao (2026)
Research skill module: Structural Approach (vs. Reduced-form Approach)

Class 9 (Nov 17): Adoption of Cleaner Technologies (Water and Sanitation)

Required readings: Kremer et al. (2011); Bennett (2012)
Optional readings: Geruso and Spears (2018); Motohashi (2026b); Cameron et al. (2022); Burlig et al. (2026)
Research skill module: Tips for Implementing RCTs

Class 10 (Nov 24): Natural Resources

Required readings: Alix-Garcia et al. (2013); Sekhri (2014)
Optional readings: Duflo and Pande (2007); Blakeslee et al. (2020); Brander and Taylor (1998)
Research skill module: Tips for Navigating Academia

Class 11-1 (Dec 1): Political Economy of the Environment

Required readings: Burgess et al. (2012); Lipscomb and Mobarak (2016)
Optional readings: Motohashi and Toya (2025); Dube and Vargas (2013)

Class 11-2 (Dec 1): Economic History of the Environment

Required readings: Alsan and Goldin (2019); Antman (2023)
Optional readings: Ambrus et al. (2020); Heblich et al. (2021); Hornbeck and Keskin (2014)
Research skill module: Tips for Accessing and Using Administrative Data

Class 12 (Dec 8): Final Presentation of Research Proposals 1

Class 13 (Dec 15): Final Presentation of Research Proposals 2

Class 14: No Class (Makeup Session if Needed)

References

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